

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	23 June 2026
Team ID	
Project Name	Software Installation Request Automation
Maximum Marks	4 Marks

Technical Architecture:

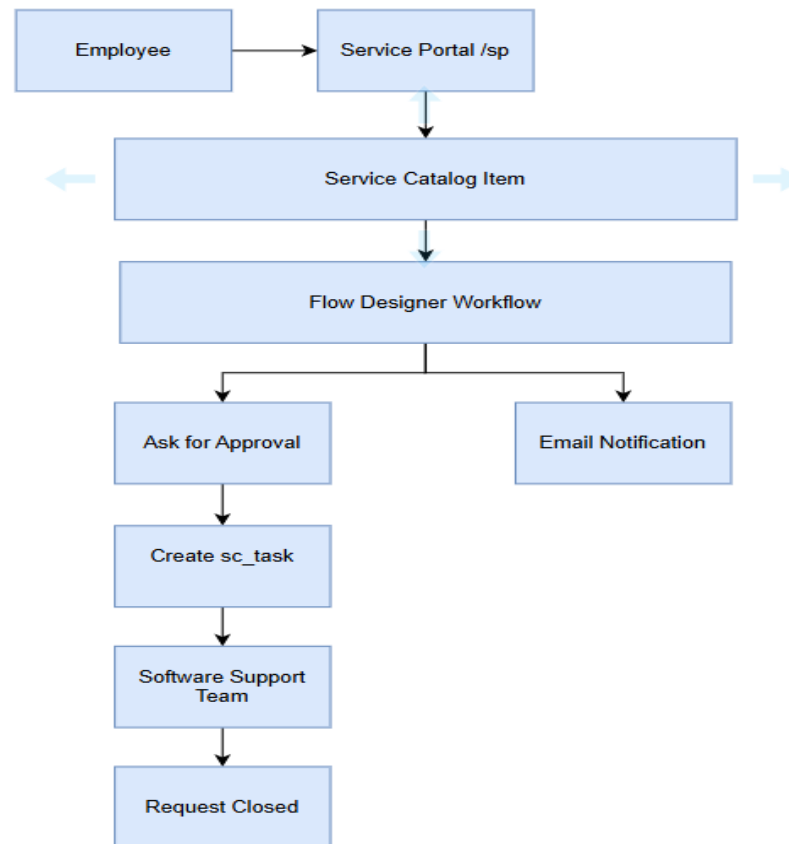


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee submits software request through Service Portal	ServiceNow Service Portal (/sp)
2.	Application Logic-1	Workflow automation for approval and task creation	ServiceNow Flow Designer (Workflow Studio)
3.	Application Logic-2	Business Rule to auto-raise incident on On Hold state	ServiceNow Business Rules (GlideRecord Script)
4.	Application Logic-3	Catalog item variables capture and validation	ServiceNow Service Catalog (sc_cat_item)
5.	Database	Stores requests, approvals, tasks and incidents	ServiceNow Tables: sc_request, sc_req_item, sc_task, sysapproval_approver
6.	Cloud Database	ServiceNow cloud-hosted instance database	ServiceNow PDI (Personal Developer Instance)
7.	File Storage	Update Set XML export for deployment	ServiceNow Update Set (XML file)
8.	External API-1	Email notification on approval or rejection	ServiceNow Email Notification Engine
9.	External API-2	Service Portal search using meta tag	ServiceNow Service Portal API
10.	Machine Learning Model	Not applicable for this project	N/A
11.	Infrastructure (Server / Cloud)	Cloud-hosted ServiceNow Developer Instance	ServiceNow PDI — cloud.service-now.com

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	ServiceNow uses JavaScript-based scripting for business rules and flow actions	JavaScript (GlideRecord, GlideSystem)
2.	Security Implementations	Only authorized managers can approve requests. Catalog item accessible only to logged-in employees. Role-based access control enforced	ServiceNow Roles & ACLs (Access Control List)
3.	Scalable Architecture	Single workflow handles all software request types. New catalog variables or approval groups can be added without redesigning the flow	ServiceNow Flow Designer — modular action-based architecture
4.	Availability	ServiceNow PDI hosted on cloud — available 24/7 for request submission and tracking via Service Portal	ServiceNow Cloud Infrastructure
5.	Performance	Workflow triggers instantly on catalog submission. Approval and task creation complete within seconds. No manual steps required	ServiceNow Async Flow Execution (background processing)